Paper 1.50 - LCR Claim Contract

CQE-paper-01.50
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-01.50.md

Paper 1.50 - LCR Claim Contract

Purpose

Paper 1.50 defines what counts as a valid claim about the LCR carrier. It is a meta-review layer between Paper 1 and the papers that depend on Paper 1.

Admitted Paper 1 Claims

The following claims are admitted from Paper 1:

```
LCR is the minimal carrier for one center and two boundary addresses.
Binary LCR has exactly eight states.
Left-right reversal preserves C.
Left-right reversal is an involution.
Binary shell counts are 1,3,3,1.
The shell-2 state (1,0,1) disproves boundary-value inequality.
The shell-2 stratum supplies the side-flip doublet interface.
```

Claim Requirements

Any later paper using LCR must state:

```
which center is being preserved
which boundary addresses are being read
which transform is applied
which Paper 1 result is being imported
whether the use is proved, bounded, candidate, or obligation
```

A later paper may not silently treat value inequality as the definition of opposition. Opposition is an address relation.

Linked Receipt

The minimum receipt link for Paper 1 is:

```
paper: CQE-paper-01
theorem: Minimal LCR carrier
receipt: production/formal-papers/CQE-paper-01/lcr_carrier_receipt.json
status: pass
```

The receipt is sufficient for the finite carrier facts. It is not sufficient by itself for the full `J_3(O)` registration or Standard Model extension claims.

Boundary Failures

The following are boundary failures:

```
claiming L and R are opposed because their values differ
using a two-address carrier while claiming one center and two boundaries
moving C without declaring a recentering
using the shell-2 doublet as a full physics proof without later-paper support
```

Boundary failures are not deleted. They are routed to obligation or later paper support.

Conclusion

Paper 1.50 lets later papers import the LCR carrier honestly. It preserves the finite proof while preventing overclaiming.